

Electricity tariff design, private investment in PV and batteries, and the distribution of impacts of tariff change

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Abstract

We study the distribution of impacts across households when dynamic electricity tariffs are introduced as a means to promote system friendly behavior of consumers (households) in Switzerland. For this we combine a building level model of rational investments in solar photovoltaics and batteries with household data for analyzing distributional effects across households. We find that the transition to dynamic tariffs somewhat favours high income households in the short term but is rather distributionally neutral in the long term when heat-pumps and battery electric vehicles (BEVs) are more widely spread. In the short term, retirees are the group that is least able to profit from dynamic tariffs; in the long term, home owners can better profit from dynamic tariffs than tenants. However, heterogeneity of impacts between households within each household group is considerable. For long term results, we can show that our results are crucially dependent on assumptions about electricity price levels and patterns.

Keywords: dynamic electricity tariffs, rational agent modeling, distributional impacts

JEL codes: D3, Q4, R2

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1 Introduction

The transition to a decarbonised energy system presents fundamental challenges to the design and operation of electricity networks. As jurisdictions around the world pursue ambitious climate targets, electricity generation is shifting to intermittent renewable sources while simultaneous electrification of transport and residential heating will substantially increase overall electricity demand. These two aspects of the transformation can create critical temporal mismatches: Renewable generation profiles increasingly diverge from traditional consumption patterns, threatening grid stability and necessitating costly infrastructure investments to accommodate potentially unresponsive peak loads.

One promising avenue for managing this mismatch without resorting to extensive expansions of the electricity system lies in exploiting demand-side flexibility. There are two mechanisms to mobilise this flexibility. Either utilities exert direct load control, or price signals facilitate indirect coordination by incentivising consumers to shift consumption away from peak periods. Time-of-use (ToU) tariffs, which vary electricity prices across predetermined time blocks, and more sophisticated dynamic pricing structures represent the latter approach, embedding system conditions directly into retail prices and allowing decentralised decision-making by households and businesses. Borenstein (2005); Faruqui and Sergici (2010) show that dynamic tariffs are effective at reducing system costs by shifting time of electricity use. Boogen and Winzer (2024) provide recent evidence, that household consumers react to price signals in Switzerland, too, and Schlecht, Ramírez-Molina, and Darudi (2025) show how this can help the cost-efficiency of the Swiss electricity sector.

While efficient, dynamic tariffs are not met with enthusiasm by consumers. This is largely due to the uncertainty that dynamic tariffs create with respect to consumers' annual electricity bill. Another disadvantage of dynamic tariffs may be that first evidence points in the direction of dynamic tariffs putting lowest-income households at a disadvantage (Borenstein, 2013). This affects the public perception and political desirability of dynamic tariffs. In the context of the Swiss electricity market, where households buy electricity from local monopolies that are often (majority) public owned, public acceptance of dynamic tariffs can influence their political practicability.

With this study we provide an in-depth analysis of the distributional effects that can be expected from the introduction of dynamic tariffs in a wealthy developed country like Switzerland for the scenario years 2020 and 2050. We run a numerical analysis that captures the interrelations of building stocks, socio-economic properties, and geography in the Swiss population to determine how opportunities to react to dynamic tariffs as well as overall exposure to the advantages and disadvantages of dynamic pricing are distributed across different parts of the Swiss population. More specifically, we employ a buildings level model of energy system optimization (SwissStore; see Schlecht, Ramírez-Molina, & Darudi, 2025) and analyze the distributional effects of tariff design changes by allocating energy system cost changes to households living in the buildings. The optimization of the energy system of buildings uses comprehensive data on Swiss building stock and high-resolution geographic data on solar irradiation and climatic conditions that affect the demand for space heating and the efficiency of heat pumps. It also uses hourly-resolution data for decision-making about the deployment of flexible assets and storage options. It assumes rational agents that take buildings' heating infrastructure as given and optimize investments in solar PV and batteries, as well as deployment of available flexibility in electricity demand and supply from and to the electricity grid to minimize the overall cost of the buildings' energy

systems while holding constant the level of energy services provided to inhabitants. For the distributional analysis on the household level, we link household level survey data to building data based on survey information about what types of dwellings households occupy.

Electricity tariffs are composed of an energy component reflecting the cost of generating electricity, a grid component that compensates utilities' cost of building and maintaining the grid, and other levies and taxes. Both the energy component and the grid component can be made dynamic. But while the energy component is made dynamic quite naturally by reflecting electricity day ahead prices, e.g., making grid tariffs dynamic is not as straightforward. In addition, recent research has shown that time-of-use grid tariffs can be unreliable in reducing maximum grid use (Ovaere & Vergouwen, 2026),¹ As a consequence, we focus on the energy component for designing temporal patterns in dynamic tariffs for this study. Anecdotal evidence from 2026 summer days in the canton of Zurich suggests that this is enough to capture relevant parts of the variability of dynamic electricity tariffs across the day.

We make the following contributions to the literature. First, we complement previous studies that compared the impacts of tariff redesign among households of different income levels. In addition, we explore inequalities across numerous additional socio-economic dimensions. Second, and most importantly, our study design makes it possible to analyze how flexibility potential is distributed across different parts of the population and shows the relevance of this for overall distributional outcomes.

We find that high-income households have a systematic advantage only in the short term. In the long run, when heat pumps and BEVs are ubiquitous, all income levels can use load flexibility to adapt to dynamic tariffs and profit from the redesign. Retired households also experience somewhat more negative impacts than their working-age counterparts in the short term, an effect that diminishes in the long run. Home owners have an advantage in profiting from dynamic tariffs compared to tenants in the long run. These trends are not very pronounced, and the wide heterogeneity of impacts that we find within all subgroups of the Swiss population may present the more fundamental obstacle to the popularity of the transition to dynamic tariffs, albeit one that can be mitigated by, e.g., opt-out options.

The remainder of the paper is structured as follows. Section 2 describes our numerical modeling framework, the data harmonization and matching procedure, and the scenarios used in the comparative static analysis. Section 3 discusses our findings in detail, and Section 4 puts our findings in perspective.

2 Method

We can expect a range of things to affect consumer welfare when electricity tariffs are switched from uniform to hourly. The first and most direct effect is that energy expenditures for given levels of energy services change. Depending on consumers use of temporal flexibility for consuming electricity this may include shifting some electricity demand to periods when electricity is cheaper. But if consumers know about tariff design beforehand, they may also change how much hardware like solar PV and batteries they install for exploiting price differences and optimize own consumption in order to avoid grid tariffs and auxiliary fees. These first order effects on energy related expenditures at fixed levels of energy service consumption can be described

¹Peak demand pricing appears to be giving households the more reliable incentive to keep the maximum load on the grid low.

by a program of cost-minimization and we use SwissStore, a buildings level model for energy cost minimization at given energy prices to implement this.

Second, two effects on the electricity market may influence market prices for electricity when consumers react to changing tariff design by reducing or increasing consumption at different points in time: On the one hand, the energy component of the electricity price may change as changes in consumption patterns may influence which technology will be the marginal, price-setting one. And on the other hand, the grid fee component of the electricity price may have to be adjusted as the same grid – or even expansions to it – may need to be financed based on different volumes of electricity consumption, depending on tariff design and consumers’ reaction to it. These effects are not explicitly modeled, but we report changes in volumes of electricity consumed from the grid to give an estimate of how much grid tariffs may have to be adjusted due to tariff design, and we observe that, according to the electricity market model FEM, cross-border transmission capacities are not fully used in most hours of 2050, which suggests that Swiss spot market prices are mostly determined by the electricity prices in the surrounding countries and thus, changes in households’ electricity consumption are unlikely to significantly influence spot market prices.

In the following, we describe the buildings level models, SwissStore (Section 2.1), and the data basis for the distributional analysis (Section 2.2).

2.1 The SwissStore model

We use the SwissStore model to evaluate how households might respond to dynamic electricity tariffs and the impact this could have on investments in solar PV and batteries. To estimate the future impact of these tariffs on the Swiss population, we have set 2020 as a benchmark to be compared with 2050. This enables us to gain a better understanding of the impact of these tariffs under various regulatory and market conditions.

The model has previously been applied in the AGGREGATE project and a description of the model is given in Appendix A of the project’s final report (Schlecht, Ramírez-Molina, & Darudi, 2025). In the following subsections, we give a short description of SwissStore and the scenario variants we simulated on this paper.

2.1.1 Model description

SwissStore is a bottom-up greenfield investment energy optimization model of the Swiss residential sector. The model integrates the electricity, heating and private electric mobility demand of households (HHs) with endogenous investments in solar PV and battery technologies. The objective function minimises the annualised total energy system costs under different policy, cost and electricity tariff scenarios. Costs include yearly energy consumption (electricity and heat) and annualised investment and operational costs.

Figure 1 shows SwissStore’s modules: the household archetype module (green), the heating technologies module (orange), and the BEVs module (blue).

The household archetype module describes the building’s physical characteristics and geographical location, as well as the technologies in which end consumers can invest after learning what the electricity tariff design is: solar PV and battery storage. These investments are constrained by the available roof surface of each building archetype ².

²We limit the maximum battery storage capacity to 150% of the maximum solar PV installed capacity,

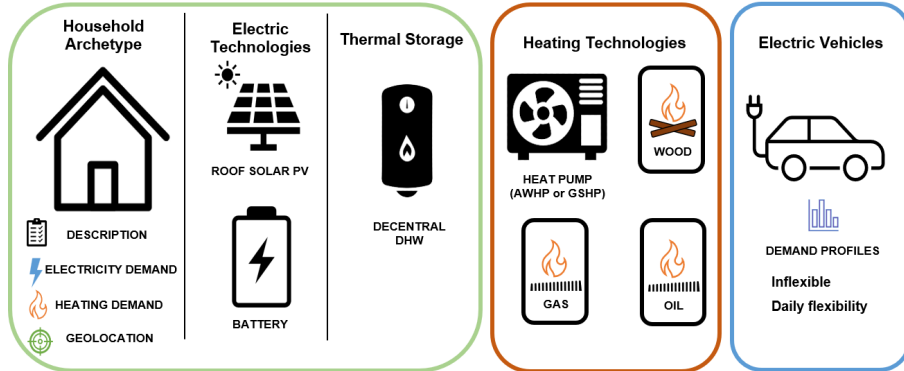


Figure 1: SwissStore's modules

The archetype description includes building-specific heating demand profiles and information on the compatibility of different heating technologies in terms of type and heating capacity. This means that, if compatible, a building archetype can be paired with more than one heating system. In this case, we can either assign a certain technology exogenously, or let the model choose the heating system that is most cost-optimal, as determined by the policy and regulatory measures being analysed.

In this application of the SwissStore model, household's ownership of their heating system and battery electric vehicle (BEV) is scenario dependent (see Section 2.3)³. For 2020, we use survey data to match a building archetype with a specific heating technology while for 2050, we allow the model to select the optimal combination, assuming that investors will act rationally. Regarding their solar PV and battery investments, households optimally size their installations in reaction to electricity tariff design.

Our modeling framework yields the yearly energy bills for each household archetype (i.e., electricity and fossil fuel expenditures). This feature allows us to compare the effect of tariff design and flexibility on the annual energy demand expenditures of the household (i.e., without considering investment and operational costs). When adding the annualized investment costs and operational costs of PV and batteries to households' expenditures, we obtain the *energy system cost* of buildings (that is, annualized cost of owning and operating the modeled energy assets). The total cost and its subcategories — namely *annualized investment costs*, *operating costs*, *electricity expenditures*, and *heating fuel expenditures* — are relevant information for our analysis of distributional effects.

SwissStore incorporates various policy and regulatory elements across modules, including customisable electricity tariff design, technology subsidies, building retrofit rates and CO₂ taxes. This, together with its modular design, enables us to evaluate the impact of a broad spectrum of policy-technology combinations within Swiss households.

which is itself limited by the suitable roof surface.

³This reflects our assumption that households choose their heating systems and vehicle drive trains based on average electricity prices compared to costs of other energy sources and do not sweat the details of electricity tariff design for these decisions.

2.1.2 SwissStore data

Building characterization and space heating demand

Data on residential buildings and their respective space heating demands are an output of the Swiss Residential Building Stock (SwissRes) model (Streicher et al., 2019). SwissRes simulates space heating demand profiles for households with a daily resolution using information on building elements and envelopes, geographical location and historical local weather conditions.

Residential buildings are depicted by representative archetypes. Each archetype represents multiple buildings that are aggregated according to relevant characteristics such as age, household type (single or multi-family household; SFH, MFH respectively), urban typology (urban, suburban, rural) and geographical location.

The characterisation of these archetypes is based on the Federal Register of Buildings and Dwellings (FRBD), a comprehensive database of all buildings in Switzerland. SwissStore’s dataset captures the totality of the Swiss housing stock, comprising more than 330,000 housing archetypes accounting for 1.4 million buildings.

Household socioeconomic data and implicit discount rates

We have expanded the scope of SwissStore to include socioeconomic data on household residents. This was achieved by matching building archetype data with responses to the Swiss Household Energy Survey (SHEDS), which surveys households with respect to their socioeconomic situation on the one hand and collects information about the respondent’s dwelling on the other (Farsi & Weber, 2024). The data matching methodology is discussed in detail in Section 2.2.

Socioeconomic data allow us to simulate different investment behaviours among households according to their income level. There is abundant literature showing that the implicit discount rate (IDR) of investments differs between socioeconomic groups, particularly inversely with income (Haq & Weiss, 2018; Hausman, 1979; Lawrance, 1991; Newell & Siikamäki, 2015; Schleich, Gassmann, Meissner, & Faure, 2023; Train, 1985). This concept has been applied to solar PV investments (Bollinger, Gillingham, & Kirkpatrick, 2025; De Groote & Verboven, 2019; Heeter, Sekar, Fekete, Shah, & Cook, 2021), building retrofits and heating system investments (Andersen, Wiese, Petrovic, & McKenna, 2020; Steinbach & Zamasz, 2017) and broader energy system model applications (García-Gusano, Espegren, Lind, & Kirkengen, 2016; Löffler, 2021; Schleich, Gassmann, Faure, & Meissner, 2016).

Based on this evidence, in this study we implemented a linearly decreasing IDR relative to income for solar PV and battery investments. Households in the richest income bracket apply an IDR equal to an assumed market rate of 4% (García-Gusano et al., 2016; Steinbach & Zamasz, 2017). We set the upper IDR limit to 15% to lowest-income households following the work of Nægeli, Jakob, Catenazzi, and Ostermeyer (2020). This value arises endogenously from their agent-based model of energy retrofit investments by Swiss households and it lies within the ranges reported in the literature, while also reflecting the Swiss residential context.

Inflexible electricity demand

Inflexible electricity demand profiles are computed according to the methodology described by Rinaldi, Yilmaz, Patel, and Parra (2022). First, we generate an estimate of the average electricity consumption per square meter (kWh/m²) for each type of

household and urban topology combination (e.g., SFH-Rural, MFH-Urban). We do this by distributing the national total electricity demand of the Swiss household sector across household types and dividing it by their total surface, using data from the FRBD database. In a second stage, normalized hourly consumption profiles are derived for each household archetype from real anonymized smart meter consumption data from various Swiss utility companies.

BEV demand

BEV demand data is sourced from Winzer, Ludwig, Auer, and Hlawatsch (2023). This dataset presents a number of BEV demand profiles with hourly resolution. In the absence of information about household-specific driving patterns, and in order to avoid unexplained heterogeneity in the results, we have assumed that all households exhibit the same driving demand, equivalent to the mean of all profiles.

Electricity price data

For the scenario year 2020, we use real electricity price data from the Swiss Federal Electricity Commission (ELCom) (Elcom, 2024) of the year 2023. We chose this year to avoid the atypical period from 2020 to 2022, which was characterised by the global pandemic of the virus SARS-CoV-2 (Covid-19) and the European energy crisis that unfolded following the Russian invasion of Ukraine.

As our analysis focuses on the residential sector, we retrieved reported residential tariffs, broken down by component (i.e. energy cost, grid charges, renewable energy support and other fees), from every utility company in the country. We then averaged the prices at the national level. As a result, we obtained hourly spot price profiles (i.e. energy cost) as well as average values for all residential electricity price components in Switzerland.

To model the year 2050, we used hourly electricity spot price time series from the Future Energy Model (FEM) (Darudi, Savelsberg, & Schlecht, 2024), an optimization model of the European electricity market. To account for future price and policy uncertainty we designed two price scenarios, differentiated by Switzerland's level of integration into the European electricity market. Assuming full market integration results in lower overall price levels (scenario *lowp*); while limiting cross-border trading to 30% produced significantly higher prices (scenario *highp*). Both scenarios are based on weather data from 1995, which is considered an average year.

2.1.3 Model variants

SwissStore allows for different assumptions about how households will deploy the technologies available to them under different tariff structures. For our analysis, we use a range of model variants to simulate household behaviour with respect to flexibility of electricity consumption (i.e., flexible heat pump and BEV operation, including charging for the latter) and investment.

The following gives a description of the different variants of the model and Table 1 summarizes and provides an overview.

Heating flexibility variants

To simulate inflexible space heating demand in the *hp_inflex* scenario, we distribute SwissRes' building-specific daily heating demand to each hour of each day throughout

the year using hourly profiles for residential buildings from Schlecht, Darudi, et al. (2025).

In the case of the *hp_daily* scenario, we allow the heat pump to meet the space heating demand within a flexibility time window of 24 hours. The source of the resulting flexibility is the thermal inertia of the building that allows flexible operation of heat pumps.

It is important to note that the flexible operation of a heat pump is constrained by its rated power capacity, which is scaled according to the maximum daily demand, in accordance with technical criteria. This means that the heat pump cannot provide the entire daily demand in one or a few time periods when electricity prices are low; instead, it must distribute the supply throughout the day.

BEV flexibility variants

To construct the BEV flexibility variants, we first aggregate the average hourly BEV demand data to obtain a new profile with daily resolution. For the *bev_inflex* variant, the model assumes that charging will start at 18:00 every day and continue until the daily demand is met. As in the *hp_daily* variant, *bev_daily* provides daily flexibility to meet energy requirements, subject to the availability of BEVs for charging. Charging power is limited to 7 kW in households, in line with the assumptions made by Parajeles Herrera and Hug (2025)

When we model the impact of BEVs on electricity demand flexibility, we have to make assumptions about the availability of BEVs at the private power outlet. Again, as we have no information about driving patterns or BEV availability at home, we model two edge cases to highlight the importance of this assumption for modeling results. In the baseline, we assume that BEVs are away during working hours, i.e., from 07.00 to 16.00 hrs (scenario variant *commute*). As the other extreme, we assume that BEVs are always at home when they are not being driven (a scenario variant styled *work-from-home* or *wfh*), for which the vehicle is only unplugged at 07.00, 11.00 and 15.00 hrs.

Connection availability is relevant not only for the periods during which BEVs can charge their batteries, but also for when they can discharge energy for self-consumption or feed electricity into the grid, depending on the variant assumption.

PV and battery investment variants

We employ a set of three PV and battery investment variants with two goals in mind. First, we aim to disentangle the effects of implementing dynamic tariffs on existing electricity demand from those of using PV and batteries for self-consumption and arbitrage. Second, we want to examine investment incentives associated with home ownership.

The *optimal distribution of investment shares* is the proportion of the total investment that building owners and tenants would have to contribute in order to maximise their individual benefit if they would sign a contract. This optimal level of investment is case-specific, as it depends on various factors; such as PV generation potential, available roof surface area, demand flexibility potential, electricity price levels and volatility, technologies' capital and operational costs, number of dwellings per building, among others.

For example, in a large apartment building, the demand would be high and the available surface area for installing PV panels would be relatively small. Therefore,

Table 1: Overview of SwissStore’s variants of how household deploy their available technologies.

Variant	Type	Acronym	Description
Heat pump flexibility	Inflexible	hp_inflex	Inflexible heat pump operation
	Flexible	hp_daily	Daily heat pump flexibility
BEV flexibility	Inflexible	bev_inflex	Inflexible BEV charging (only unidirectional)
	Flexible	bev_daily	Daily BEV charging flexibility (uni & bidirectional)
BEV connection availability	Commute	commute	BEV is not available during work hours
	Work-from-home	wfh	BEV is available for most of the day
PV and battery investment	Disabled investments	noPB	Disabled PV and battery investments
	Only owners invest	ownerPB	Only home owners invest
	Optimal investment	optPB	Optimal investment due to contractual agreement between tenants and owners

most of the realised savings would come from the tenants' demand flexibility, which responds to the tariff design and is mostly independent of PV and battery investments if they are relatively small in scale compared to the overall demand. This means that tenants would give away most of their benefits to the owner in exchange for a small investment, therefore they would be better off assuming a larger share of the investment costs and keeping the savings derived from their change in consumption behaviour for themselves. In the case of a small building with relatively high PV generation potential (i.e., relatively large available roof surface, high generation potential, or both), this effect is minimised and grid injection and arbitrage represent a higher proportion of the benefits, meaning that the *optimal distribution share* would be more evenly distributed. Importantly, given our assumption of incentive alignment, this dynamic does not affect the optimal absolute level of investments; it only affects the distribution share of costs and benefits defined in the contract between tenants and owners.

Since determining the *optimal distribution of investment shares* endogenously is beyond the scope of this study, we conducted a sensitivity analysis with varying degrees of cost and benefit distribution shares, which we have called *noPB* and *ownerPB* (results presented in Section 3.3).

In the *noPB* variant, all investments in PV and batteries are disabled. This means that any observed difference between the scenarios presented in Section 2.3.1 would be due to exposure to dynamic tariffs and the corresponding end consumers' demand flexibility response using other existing infrastructure (heat pumps and BEVs).

In the *ownerPB* variant, investments are only considered in dwellings occupied by the owner. In this variant, we assume that misaligned incentives between building owners and tenants result in a sub-optimal outcome, whereby building owners decide not to invest at all if they are renting their property.

On the other hand, in the *optPB* variant, we simulate a contract between owners and tenants, in which they share the costs and benefits from PV and battery investments. This guarantees that all relevant agents receive full investment incentives (i.e., self-consumption, flexibility, arbitrage and generation), enabling them to rationally choose the optimal level of investment.

2.2 SwissStore model results in the distributional analysis

For a household-level distributional analysis based on the buildings-level outcomes according to the SwissStore model, we need to identify which respondents from household survey data tend to live in which buildings from the buildings registry data. Here we discuss the data requirements, which data sets cover the necessary data, and how they have been combined for the final analysis.

The central data set for our distributional analysis is SHEDS, a survey that collects information about households' socio-economic situation on the one hand and about the respondent's dwelling on the other (Farsi & Weber, 2024). We use annual waves 2017–2021 where all information needed for linking with buildings data from the buildings registry is consistently available. The matching of household observations in SHEDS with buildings in the registry data has been carried out based on the following common variables

- Building age
- Technology of the reported heating system
- Degree of urbanisation (urban / peri-urban / rural)

- Building type (single family home (SFH) / multi family home (MFH))
- Canton of residence.

In addition to these variables, a close match of dwelling size (floor area) was pursued. In SHEDS, dwelling size per household is given, in SwissStore we estimate the same by dividing building level living area by an estimate of number of apartments within the building.

We found that for some observations in SHEDS the dwelling properties do not conform with any of the building archetypes of SwissStore. When reporting modeling results of Swisstrisbution, we make sure to show results of these households (they do not experience any direct changes via SwissStore from a switch in tariff design) separately from matched households.

Since SHEDS is not a representative survey, we match SHEDS observations with observations in the Swiss Household Budget Survey “Haushaltsbudgeterhebung (HABE)” according to the statistical matching procedure previously performed by Torné and Trutnevyte (2024). We use a combination of the three waves from years 2015–2017. HABE’s observations are complemented with statistical weights that account for sampling bias and are calibrated to the observed distribution of the Swiss population (Bundesamt für Statistik, 2014; Schweingruber, Epple, Oetliker, & Rochat, 2007). For some variable combinations, there are more SHEDS observations than HABE observations and for other combinations it is the other way round. In the first case, HABE observations are allocated to multiple SHEDS observations and the statistical weights from HABE are adjusted accordingly. In the the second case, SHEDS observations (along with their corresponding SwissStore scenario results) are assigned to multiple HABE observations for the evaluation of distributional impacts in SwissStore. This results in 9855 different combinations of HABE expenditure budgets with SwissStore scenario outcomes, correctly weighted according to the population shares they represent in the population of Switzerland except for the canton of Ticino which is not covered by SHEDS.

2.3 Scenarios

SwissStore, FEM, and Swisstrisbution need to be set up in a way that can congruently represent the situation of buildings, households, and the economy at large in 2020 and 2050. For SwissStore this involves determining buildings’ equipment with heating technologies and BEV ownership, as well as the electricity and fuel prices that prosumer⁴ households are confronted with.

To build a consistent set of model assumptions, we use the same input data sources as FEM regarding carbon taxes, technology, and fuel cost projections. In general, we use scenarios from the Energy Perspectives 2050+ (EP2050+) (Bundesamt für Energie BFE, 2020), and depending on the policy scenario, we add different levels of carbon tax and fossil fuel prices to the total cost of fossil fuels, thus increasing total system costs. The results shown in this paper are compatible with the *ZERO* scenario of the EP2050+. Regarding PV and battery costs, we use data from Mellot, Moretti, Tröndle, and Patt (2024).

The stock of BEV ownership and installed heating technologies can be observed in SHEDS for years close to 2020, but has to be projected to 2050. We assume that in 2050,

⁴“Prosumer” being, of course, a portmanteau of consumer and producer. The expression reflects the fact that the same entities may demand or supply electricity from or to the grid depending on time of day and depending on electricity prices.

all households that state *car* ownership in the SHEDS survey own at least one BEV. Installed heating technologies are taken from the buildings registry for 2020. For 2050, we assume building owners to have installed the cost-optimal among a set of climate friendly options according to SwissStore in buildings that have not yet installed heat pumps or district heating.

For Swisstristribution, we employ the SWEET programme’s most recent set of common scenarios of energy-economic developments in Switzerland. These have been established by the SWEET-CROSS project and are available on the project’s website (Marcucci, 2022). Their description is given by Marcucci, Dujardin, Heinisch, Panos, and Yilmaz (2022). More specifically, we derive projections for GDP and consumption of energy carriers gas, edt, rpt, rph, and rpo from the CROSS reference scenarios. We calibrate Swisstristribution to GDP projections by scaling up the availability production factors labour and capital along with trade balances and fixed demands like government and investment spending. Consumption of energy carriers and their prices are calibrated by iteratively adjusting all sectors’ production functions until the projected consumption levels are met.

2.3.1 Tariff policy scenarios

The fundamental policy analysis in this paper is a comparison of building and household level outcomes under a flat electricity tariff and a dynamic one. The evaluation of results takes the flat tariff to be the “business-as-usual” (BaU) case and displays the differences between dynamic tariffs and this BaU as the policy impact of switching from one to the other.

Tariff construction

We assume that a liberalized electricity market will be in place in 2050 and will consequently lead to uniform prices (i.e., for the energy component of the final price) in Switzerland. Regarding the other price components, as it is unclear what pricing will be applied to grid charges and how liberalization will affect utility fees and therefore, we are taking a conservative approach by assuming the same values as in 2023.

We apply different methodologies to construct the electricity tariffs for both scenario years using the spot price data described previously for both scenario years.

Dynamic tariffs (p_hourly): Under this tariff design, end consumers are fully exposed to time-varying price signals, with both the retail price and the feed-in payment following the hourly variations in day-ahead market prices. This type of tariff gives households a strong economic incentive to manage their energy usage flexibly and maximize revenue by investing in photovoltaics and batteries to maximize electricity production and profit from arbitrage opportunities.

One of the price components is the charge to fund the feed-in remuneration programme (Kostendeckende Einspeisevergütung, KEV). In place from 2009 until the end of 2022, this programme was the main instrument used to incentivise the installation of renewable electricity generation in Switzerland, providing long-term fixed payments for electricity injected into the grid from newly built renewable power plants (Bundesamt für Energie BFE, 2026). Despite support for new plants ending with the programme, the KEV charge continues to be levied to fulfil the programme’s obligations to participating power plants.

As shown in Equation (1), the retail price is then equal to the sum of the spot price and the average national values for grid fees, KEV charge and other charges from

the utility companies, for each hour t . We assume that support for renewable energy (KEV), accounting for 2.3 *Rp./kWh* in 2023, will no longer be charged in 2050.

$$\text{hourly_retail_price}_t = \text{spot_price}_t + \text{grid_fee}_{avg} + \text{kev}_{avg} + \text{other_charges}_{avg} \quad (1)$$

Equation (2) defines the payment made to households for feeding electricity into the grid from their solar PV installations, batteries, or BEVs. As we are assuming new installations, they will not be eligible for feed-in remuneration from the KEV programme and therefore, the feed-in price they receive is equal to the hourly market energy price.

$$\text{hourly_feed_in_price}_t = \text{spot_price}_t \quad (2)$$

Flat tariffs (*p_flat*): With this tariff, households pay the same price for electricity at all times of the day and year. Under this tariff, households do not receive remuneration for feeding electricity into the grid, meaning that all PV and battery investments are made exclusively to maximise self-consumption. As the price remains constant, there is no economic incentive for households to operate their energy assets flexibly.

When designing this tariff, we first need to establish revenue-neutral tariffs for utility companies. To this end, we scale the *p_flat* tariff scenario to guarantee that, in expectation, utilities are indifferent between offering flat or dynamic tariffs. To achieve this, we first run SwissStore for a scenario in which households are exposed to dynamic prices (for both *lowp* and *highp* scenarios), their demand is completely inflexible and investments in PV and batteries are disabled (the *ev-daily-h-daily-noPB* scenario). Secondly, we use the HABE dataset to weight these results and estimate the aggregated grid demand (*tot_spot_cost*) and electricity spot costs (*tot_spot_cost*) for the entire Swiss residential building stock. Finally, we calculate the indifference spot price by dividing these values, as shown in Equation (3).

$$\text{spot_price}_{indiff} = \frac{\text{tot_spot_cost}}{\text{tot_grid_dem}} \quad (3)$$

Table 2 shows the revenue-neutral prices for the different price variants.

Table 2: Revenue-neutral flat electricity prices.

Price variant	Year	Energy price component (Rp./kWh)	Full retail price (Rp./kWh)
Uniform CH price	2020	13.97	27.92
Low price (<i>lowp</i>)	2050	5.94	19.89
High price (<i>highp</i>)	2050	25.39	39.34

The revenue-neutral flat price is then the indifference flat spot price plus the annual average of all additional cost components. This price is then applied to all hours of the year t , as depicted Equation (4).

$$\text{flat_price}_t = \text{spot_price}_{indiff} + \text{grid_fee}_{avg} + \text{kev}_{avg} + \text{other_charges}_{avg} \quad (4)$$

2.3.2 Sensitivity analysis

The results of the analysis depend on a range of assumptions that are not easily settled on. Our baseline assumption is that households are equipped with the means and willing to use heat-pumps and BEVs to pre- or postpone electricity consumption for minimizing energy system costs (scenario variants *flexible*, using flexibility assumptions *hp_daily* and *bev_daily*) but do not build batteries or photovoltaics (*noPB*). But households may fail to install soft- or hardware for exploiting this flexibility or may fail to make use of it. What changes in household’s outcomes does this alternative assumption entail? In order to analyze this, we employ a scenario variant *inflex*, where buildings’ demand profiles are completely fixed (flexibility assumptions *hp_inflex* and *bev_inflex*). In order to see what the benefits of investing in batteries and solar photovoltaics are, we further distinguish between the cases where households only use heating and BEVs flexibly but do not invest (*flexible*) and where households additionally invest in batteries and photovoltaics (*flexible+invest*). Table 3 shows a description of these scenarios.

Table 3: Modeled flexibility scenarios. If not specified otherwise, BEV availability for home charging is according to *commute*, and 2050 prices are according to *lowp*

Flexibility scenario	Combination of model variants	Description
<i>inflex</i>	<i>hp_inflex</i> - <i>bev_inflex</i> - <i>noPB</i>	Inflexible demand without investments
<i>flexible</i>	<i>hp_daily</i> - <i>bev_daily</i> - <i>noPB</i>	Flexible demand without investments
<i>flexible+owners_invest</i>	<i>hp_daily</i> - <i>bev_daily</i> - <i>ownerPB</i>	Flexible demand with PB investments by home owners only
<i>flexible+invest</i>	<i>hp_daily</i> - <i>bev_daily</i> - <i>optPB</i>	Flexible demand with optimal investments

Regarding our BEV availability assumptions, for the purposes of this study we adopt a conservative approach by focusing only on the *commute* variant in order to avoid overestimating the flexibility potential of Swiss households in our main analysis. We show what would happen if all BEV owners had BEVs available for flexible charging throughout the day (*wfh* variant) in a separate sensitivity analysis.

3 Results

Comparing different tariff designs under different scenarios, gives insights into how technology deployment changes at the building level and how this impacts consumer utility on the households. The former is modelled by the SwissStore model and analysed at the building stock level in Section 3.1. The latter is assessed within the household level distributional analysis the results of which are discussed in Section 3.2.

3.1 Building level investments and costs

PV and battery investments

Households investment incentives for PV and battery technologies are sensitive to our regulatory scenarios and price and cost assumptions. For analysing the dynamics of different tariff designs (i.e., *p_flat* and *p_hourly*), BEV demand flexibility, and households' income level, we use the flexibility scenario *flexible+invest* (Table 3).

Figure 2 and Figure 3 illustrate the total PV and battery installed capacity for all modeled households in 2020 and 2050. The figures portray the results by income class, electricity tariff scenario and BEV flexibility variants. Both figures show results for *unidirectional* BEV charging with a *commuter* driving profile and the *hp_daily* heat pump flexibility variant. For 2050, the middle panel shows the results for the *lowp* scenario, and the bottom panel shows the results for the *highp* scenario.

We observe similar trends for both solar PV and batteries. In 2020, PV capacity deployment was low, with most of it concentrated in the top three income brackets and no battery investments. By 2050, both the *lowp* and *highp* scenarios show significantly higher deployment of PV and batteries than in 2020, with higher investment levels in the *highp* scenario across income classes.

We can also observe the difference in investments between flat or dynamic tariffs. Exposure to dynamic tariffs leads to higher overall investment. Given the absence of grid-injection revenue under flat tariffs, installed capacities in this case are optimised to maximise household self-consumption, whereas the additional capacity observed with dynamic tariffs maximises revenue from grid injection and energy arbitrage using batteries.

This effect is primarily explained by two factors: lower investment costs in 2050 and the overall level of electricity prices. This interaction can be seen by comparing the two 2050 scenarios, which only differ in terms of electricity prices. Higher electricity prices lead to greater deployment because it becomes more profitable for households to maximise their self-consumption and electricity revenues when market prices are high. Furthermore, in this scenario, prices are high enough for households in lower income brackets to invest, despite them facing considerably larger IDRs than high-income households. In contrast, when we compare the year 2020 with the low price 2050 scenario, the deployment is higher in 2050 despite the fact that overall electricity prices are lower than in 2020 (see Table 2). In this case, however, it is lower technology capital costs driving higher investments.

Another explanatory factor for investment heterogeneity per tariff and income class is the amount of suitable roof area available for solar PV installations. Figure 4 shows the proportion of PV-covered roof surface area for each scenario variant. The difference in roof coverage between the *p_flat* and *p_hourly* tariff scenarios results from households profiting from the injection of electricity into the grid. A combination of dynamic tariffs (*p_hourly*) with high electricity prices *highp* leads to almost full deployment across all income brackets (the two lowest reach 99%), meaning that deployment levels shown in Figure 2 for this scenario configuration are the maximum possible, with high electricity spot prices being the main driver for PV and battery deployment. Therefore, low deployment levels in the three lowest income classes are constrained not only by facing higher IDRs, but also by lower roof surface area availability⁵. Under a flat tariff (*p_flat*) the share of roof coverage is also uniform across

⁵According to our results, the aggregated PV-suitable roof area for the lowest income bracket is 7.45 km², compared to 39.63 km² for the highest bracket. This is because richer households tend to live in single-

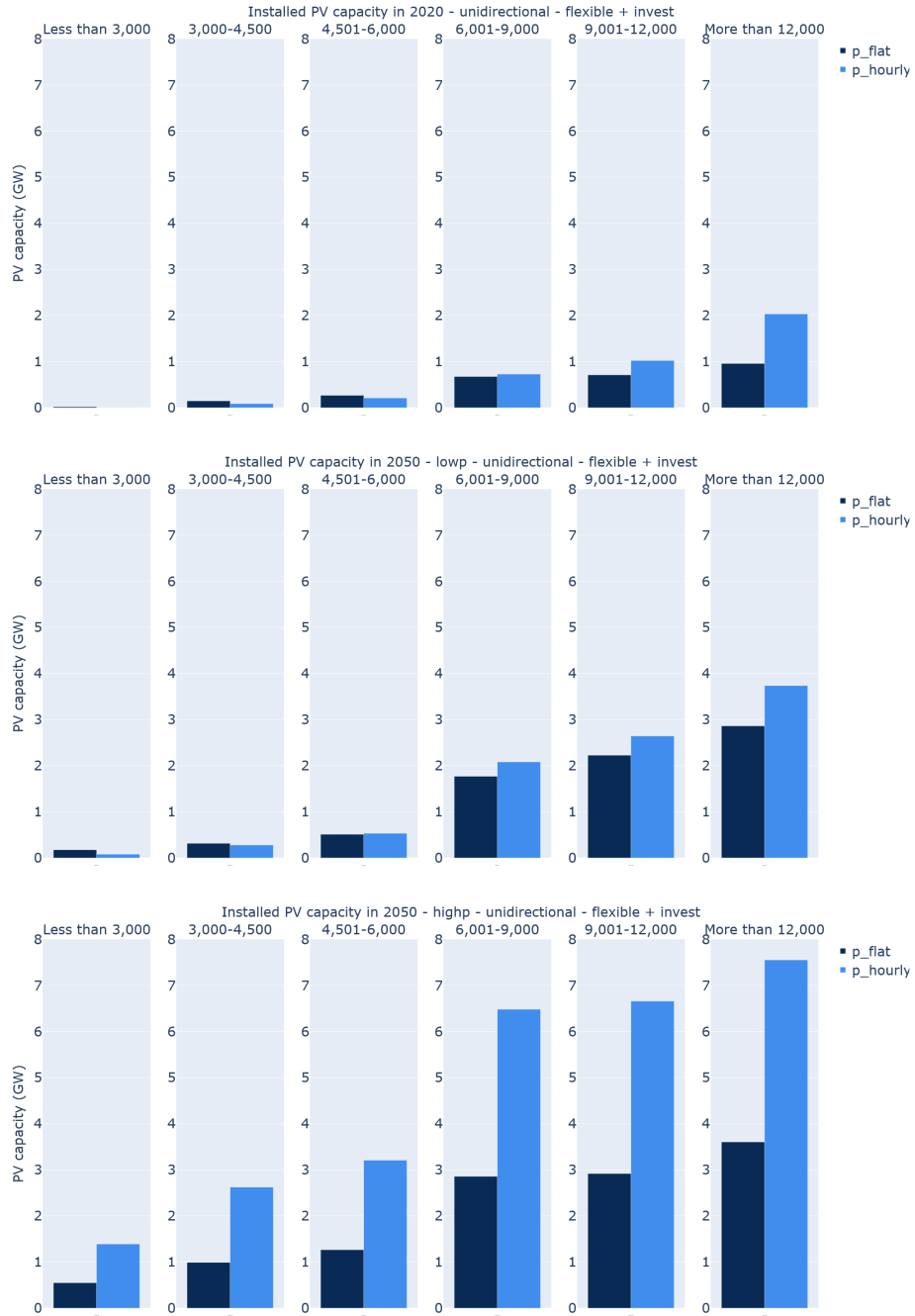


Figure 2: Total installed PV capacity (kW) per building and income class in 2050. Top panel shows results for 2020. 2050 results are shown in the middle (low price scenario) and bottom panel (high price scenario).

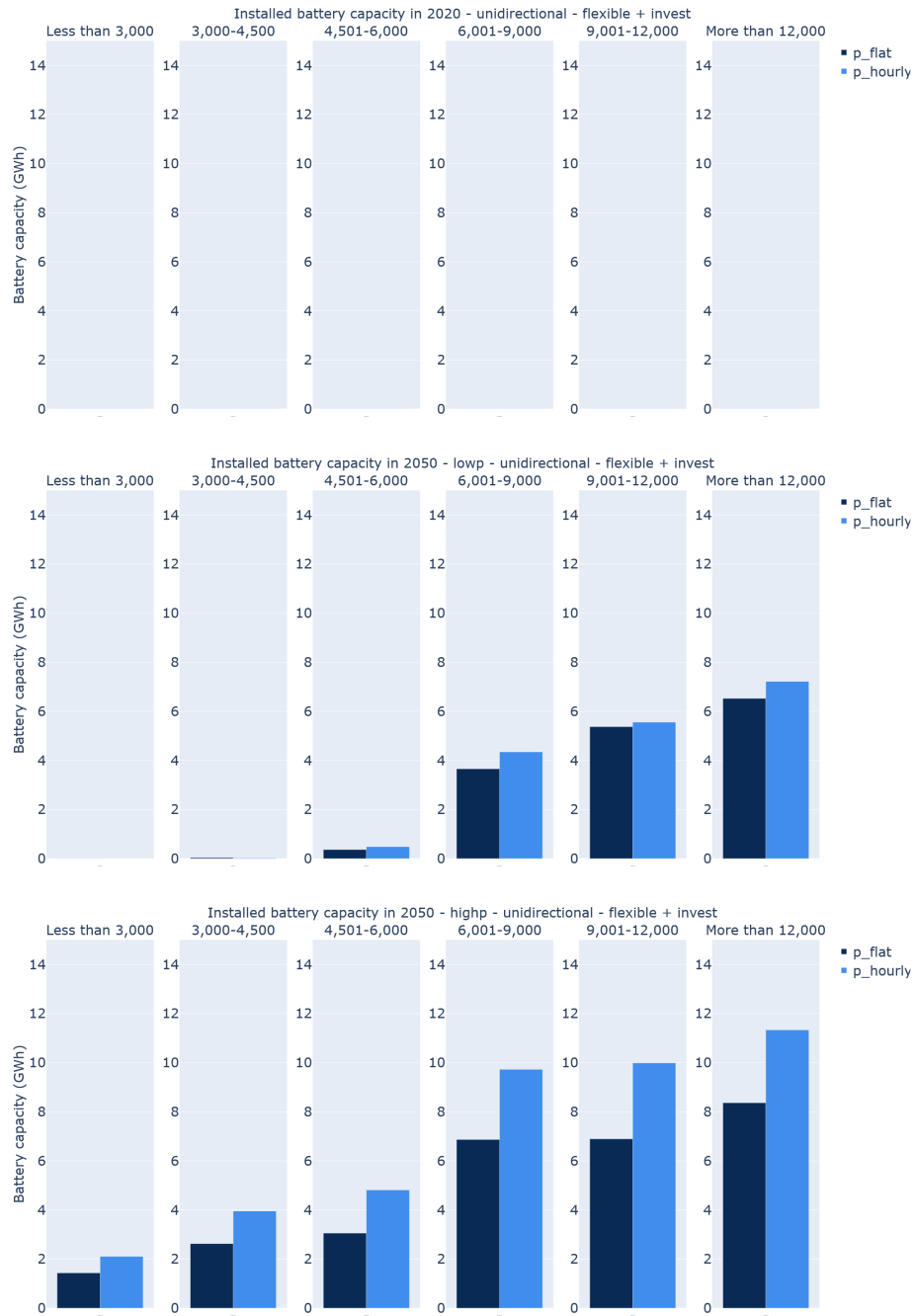


Figure 3: Total installed battery capacity (kWh) per building and income class. Top panel shows results for 2020. 2050 results are shown in the middle (low price scenario) and bottom panel (high price scenario).

income brackets, indicating that all households maximised self-consumption when facing high electricity retail prices.

In contrast, the difference in roof coverage between flat and dynamic tariffs is smaller in the *lowp* scenario than in the *highp* scenario, as lower electricity prices provide households with fewer incentives to sell electricity. In this setting, we also identify the effect of households benefiting from flexible energy asset usage, particularly in the two lowest income brackets. We observe that households invest less with dynamic tariffs than with flat tariffs, i.e. less than the optimal level of self-consumption under a flat tariff. This means that demand flexibility enables them to reduce their investments by allocating their energy usage to low-price periods and/or selling energy during high-price peaks. Additionally, we can infer that their investments do not exceed these levels due to low electricity prices and being constrained by higher IDRs.

family homes, which have a larger roof area per dwelling, unlike multi-family buildings where many dwellings share a relatively small roof area.

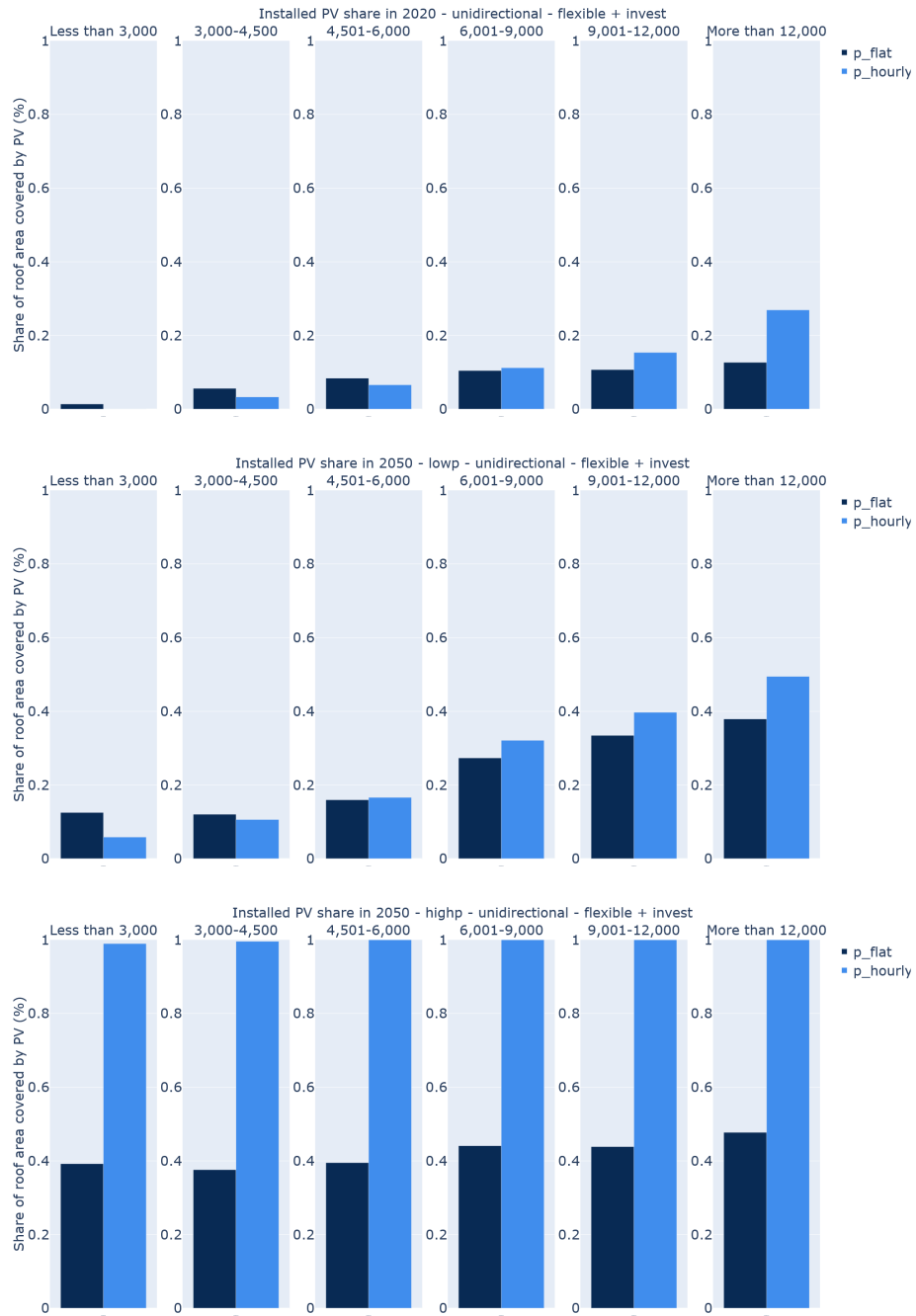


Figure 4: Share of PV-covered roof surface area (%) and income class. Top panel shows results for 2020. 2050 results are shown in the middle (low price scenario) and bottom panel (high price scenario).

Aggregate results

Figure 5 shows the total energy system cost of the entire Swiss residential sector under various flexibility settings (red bars). These results are statistically representative of the entire Swiss residential sector and were derived by allocating SwissStore’s results on costs and expenditures to households living in those buildings according to our statistical matching procedure. We observe that with the limited deployment of solar PV and batteries in 2020, there is not a highly significant difference when applying dynamic or flat tariffs. However, all scenarios show a general trend towards lower total energy costs when combining demand flexibility with dynamic tariffs, which is also true for both price scenarios in 2050. This difference is more dramatic in the *highp* scenario in 2050, due to the higher deployment of solar PV and batteries incentivised by price levels and energy arbitrage.

Aggregated residential electricity consumption is presented in blue in Figure 5. Our simulation’s consumption levels shown for 2020 are more or less consistent with historical household electricity consumption data. According to the Swiss Federal Office of Energy, in 2020 the residential demand was equal to 19.3 TWh Bundesamt für Energie BFE (2021).

Net residential electricity expenditure is the aggregated monetary balance considering both the cost of electricity from the grid and revenues from injecting electricity into the grid. As we observe on the yellow bars of Figure 5, energy expenditure declines when households behave flexibly and are exposed to dynamic tariffs. This is related to higher PV and battery deployment, particularly in the high price scenario. These decreases in expenditures only partly translate to lower energy system costs, shown in red, as the observed lower net grid consumption and expenditures are compensated by investment costs that are necessary for maximizing households’ self-consumption of electricity and revenues from generation. In 2050 under the high price scenario with dynamic tariffs, households can become net generators, with a net injection of 15.8 TWh and total revenues of 2 billion Swiss Francs.

Net electricity consumption from the grid is a good measure of how dependent on centrally generated electricity the residential sector remains under different scenarios. At the same time the grid is financed on fees levied on the total consumption of electricity from the grid but not from feeding electricity *to* the grid. To give a measure of how much this tax base for financing the Swiss electricity grid changes, we compute the percentage changes of electricity consumed from the grid under the *flexible+invest* scenario when we transition from flat electricity tariffs to dynamic ones. Our simulations indicate that in a 2020 setting, electricity consumption volumes would decrease by 3.5 percent, in the low-price scenario for 2050 by 4.8 percent and in the high-price scenario by 27.4 percent.

3.2 Distribution of household level outcomes

Households, under the switch from flat to dynamic tariffs, will experience direct impacts on the expenditure side from adapting their system investments, as well as electricity costs at constant levels of energy service provision.

To show how impacts are distributed across the Swiss population, we group Swiss households according to several socio-economic indicators (such as income, age, and home ownership). To capture the heterogeneity of impacts within these groups, we show the results as box plots. The boxes drawn in these plots are characterized by three horizontal lines: the values of the y-axis for the observations in the first, second,

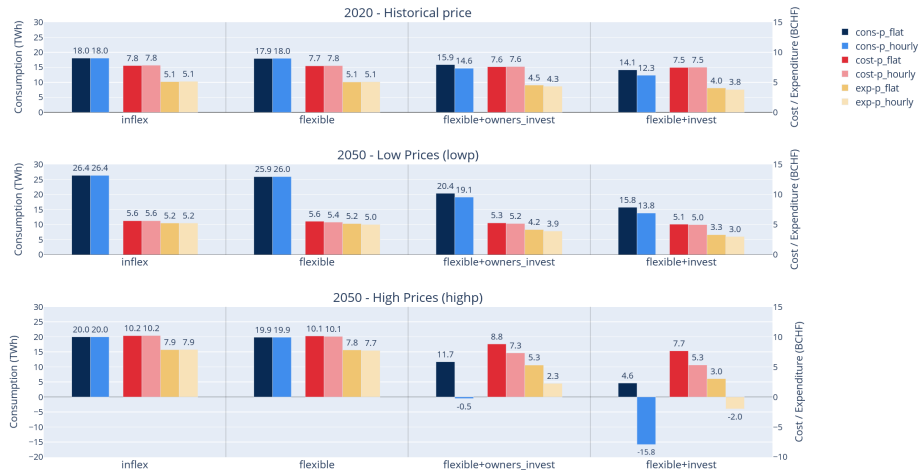


Figure 5: Total modeled residential electricity consumption (blue; in TWh), aggregated residential energy system costs in Switzerland (red; in billions CHF), and electricity expenditure (yellow; in billions CHF). Top panels show results for 2020. 2050 results are shown in the middle (low price scenario) and bottom panels (high price scenario)

and third quartiles. The boxes are complemented by whiskers that reach out to the last observation that lies less than 1.5 times the interquartile range (the distance between the first and the third quartile) beyond the box.⁶ Krzywinski and Altman (2014) give a more in-depth discussion of box plots and whiskers and how they relate to underlying distributions of observations.

Figure 6 shows changes in the annual cost of the household energy system in CHF for different assumptions about how well households can respond to dynamic tariffs. The top row of panels shows the situation at 2020 prices and building infrastructure. If households cannot respond at all (*inflex*), the fact that dynamic prices are calibrated to generate about the same revenue as flat tariffs means that, on average, households have almost the same electricity cost under both tariff designs, and the difference in energy costs is, again on average, close to zero. Yet different households have different temporal patterns to their load curves and individual households will experience changes in their electricity expenditures leading to the patterns in panel A.⁷ High-income households with higher energy spending experience higher changes in costs from the switch to dynamic tariffs and the variation of impacts within this income group is large compared to differences between income groups. When households use the flexibility of already installed equipment such as heat-pumps and BEVs (*flexible*), they have options for cost savings and they can exploit the dynamic tariffs to their advantage leading to results in the middle column of panels. Panel B shows that, in 2020, these options of cost savings are limited and the distribution of impacts from the tariff design switch look similar to panel A. Exceptions are households in the highest income quintile group, where BEV adoption is highest. Panel C, finally, shows the

⁶It is customary to plot outliers beyond the whiskers as explicit dots, but we abstain from showing them here, directing the attention to the main bulk of the 'observations rather than to single observations.

⁷SwissStore's assumption that building-level load curves only depend on building type, heating system, and BEV ownership likely underestimates the heterogeneity in household level load curves.

cost differences when households invest in solar PV and batteries to optimize their own consumption of PV generated electricity and avoid high price hours of the day (*flexible+invest*). It turns out, that this strategy works better under a flat electricity tariff than under a dynamic one and investors' total energy system costs tend to rise when dynamic tariffs are introduced.

In 2050 the situation looks different in the low-price scenario (second row of panels in Fig. 6). With a high market penetration of heat-pumps and BEVs, all households from all income classes have opportunities to react to the dynamic tariffs that we calibrate to FEM model results. Cost savings can be achieved for most households, but high-income households can save larger amounts both without (panel E, showing *flexible*) and with (panel F, showing *flexible+invest*) the options of installing solar PV and batteries available. We see that the *reduction* in energy system costs from the tariff switch is smaller if the option of installing solar PV and batteries is available than if it is not (this is clearly visible for income quintile groups 2–4). This is due to the fact that, with PV and battery installations being available, the flat tariff scenario already contains considerable energy system cost savings from these installations (see Fig. 5).

In the high-electricity price scenario (*highp*), overall building energy system costs increase considerably, which is in part due to an increase in electricity expenditures (bottom panels in Fig. 5), but another component of the energy system cost increase comes from switching away from electricity intensive heating systems. With high electricity prices, the cost savings from investing in solar PV and batteries increase considerably and these investments also yield the substantial advantage of dynamic tariffs in Panel I (bottom row of panels in Fig. 5 and 6). Since high-income households have the better opportunities to exploit solar PV with batteries, the results are more skewed towards favoring high-income households than in the low-price scenario (compare panels F and I).

Next, we show how these absolute cost numbers translate to percentage changes in the utility of consumption (see Figure 7). First, we divide absolute costs by households' baseline expenditure budgets from HABE to arrive at a relative cost difference. While considering absolute numbers reveals how large changes in households' budgets are, the relevance of these changes is not immediately clear, since we do not know how much the changes affect the consumption habits of said households. Relative numbers improve on this by taking into account the spending power of households and showing the *relative* impacts of changes in energy system costs on consumption utility. Figure 7 shows the absolute cost changes in the left-most panels and compares them to the relative cost reductions (or consumption utility increases) in the middle panels.

The comparison reveals that, because they have different consumption budgets, low and high-income households experience quite similar relative cost impacts by the transition from the flat to the dynamic electricity tariff in 2020. In 2050 and assuming the low electricity scenario, most households in all income groups can profit from cost savings, and although high-income households are better off in absolute terms, the positive relative impacts are similarly important for low-income households in relative terms. If flexibility in operating existing assets like heat-pumps and BEVs is widespread as in our scenario for 2050, most low-income households can profit from relatively high cost-savings in a transition from flat to dynamic electricity tariffs. It should be noted that the heterogeneity of impacts is large and all income groups contain at least some households that lose and some that gain under all scenarios.

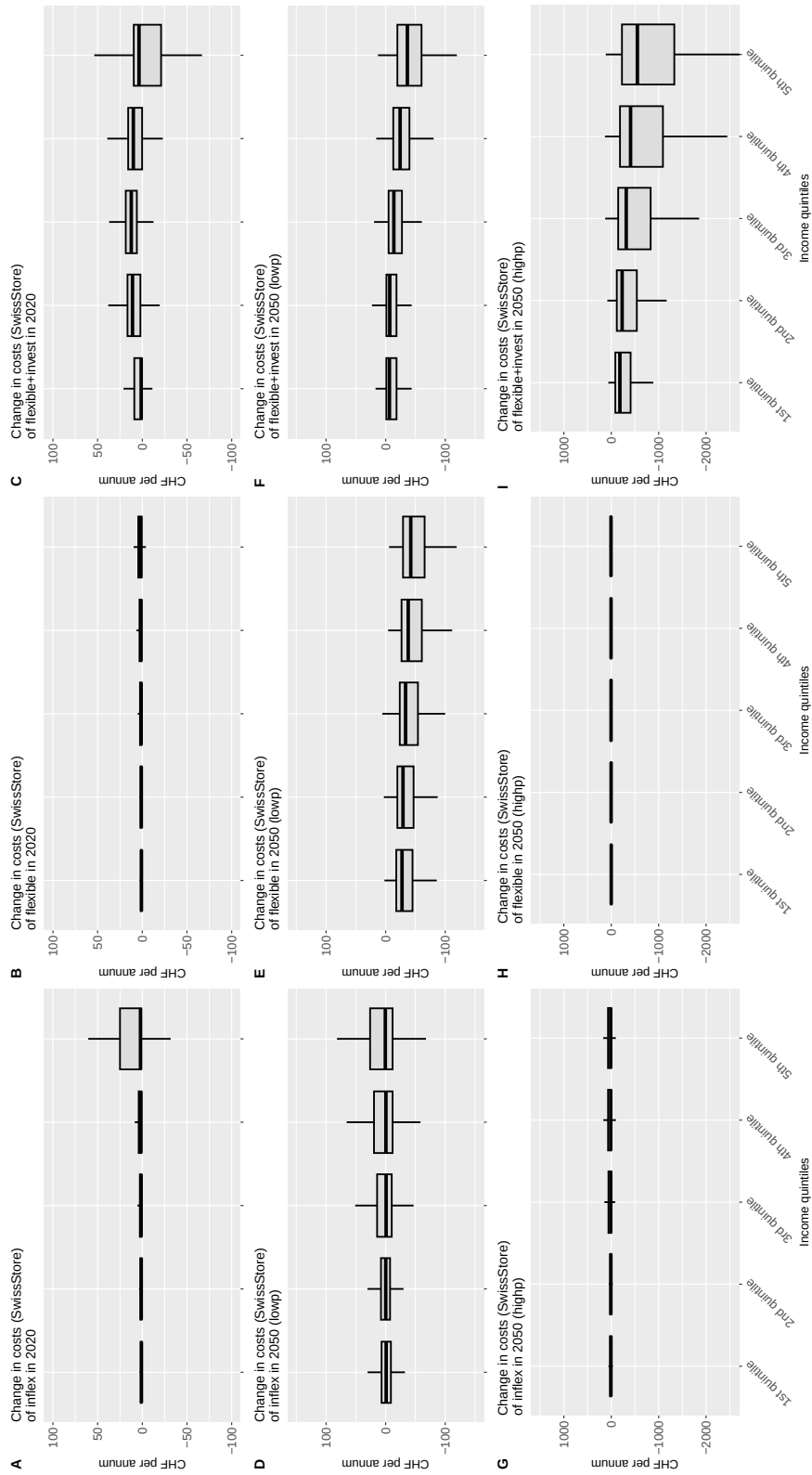


Figure 6: Distribution of changes in households' energy cost in 2020 (panels A–C) and 2050 in the *lowp* scenario (panels D–F) and in the *highp* scenario (panels G–I). Panels in different columns show different degrees of adaptation of households to tariff design: the *inflex* scenario results are shown in the left column, the *flexible* scenario in the middle column, and the *flexible+invest* scenario in the third column.

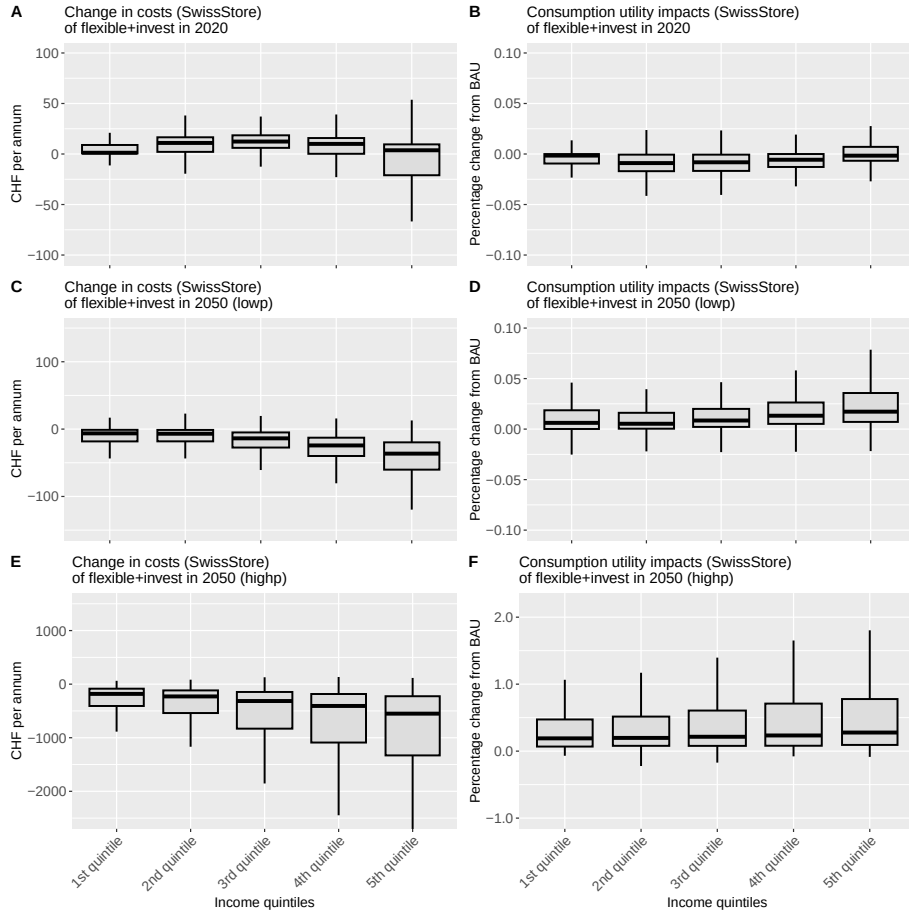


Figure 7: Household specific outcomes for the *flexible+invest* case in 2020 (panels A–B), in the *lowp* scenario for 2050 (panels C–D), and in the *highp* scenario (panels E–F). Panels left show cost differences, and in the right column, they show relative cost reductions (negative cost differences) compared to baseline expenditure budgets.

Distribution across household groups other than income quintiles

Our model setup allows us to analyze distributional impacts not just with respect to income groups but with respect to other socio-economic indicators as well. The left hand side of Figure 8 shows that the tariff switch hurts retired households more than working age households in 2020. In 2050, under the low-price scenario, retired households live in buildings that actually let them exploit dynamic tariffs well, which gives them an advantage in terms of direct costs (panel B). For the high-price scenario in 2050, retired households and those of working-age are on a similar footing in terms of direct costs (panel C). This modeling result is explained by the properties of the buildings in which these households tend to live and by internal discount rates that we model to depend on income levels. It is not dependent on elderly people's reluctance to invest in new technologies or their tendency to consume electricity at different times of the day, since SwissStore does not include such information.

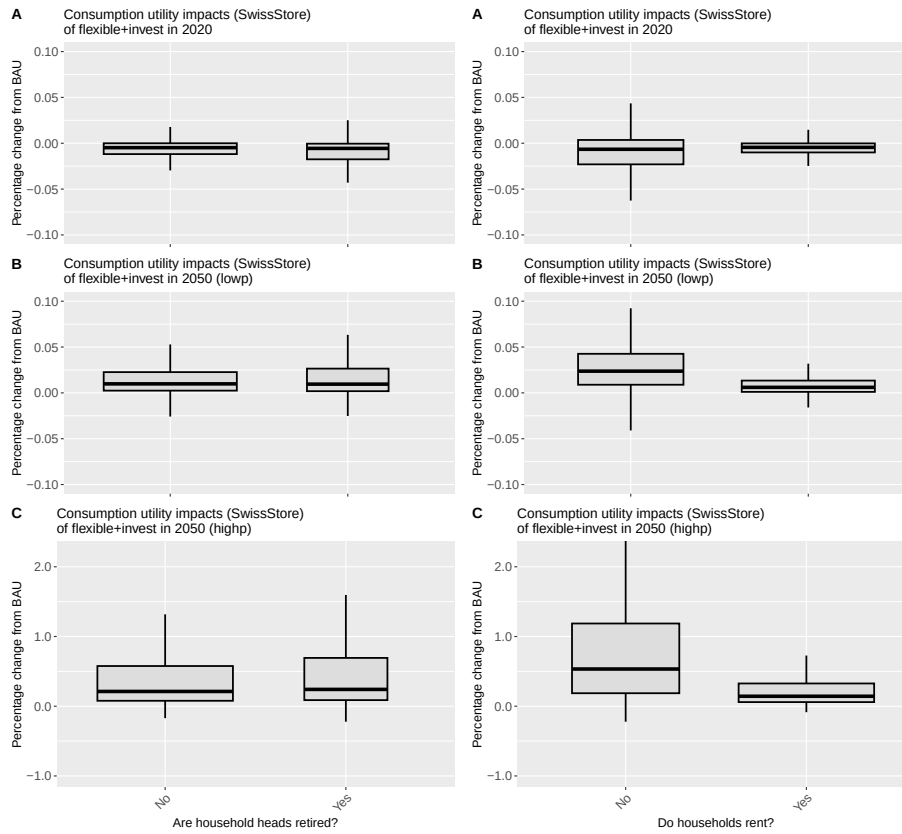


Figure 8: Left: Household specific outcomes in the *flexible+invest* case differentiated with respect to age for 2020 (panels A) and for the *lowp* scenario of 2050 (panels B), and for the *highp* scenario (panel C). Right: Household specific outcomes differentiated with respect to home ownership in 2020 (panel A) and 2050 in a low price scenario (panel B) and in a high price scenario (panel C). Panels show relative cost reductions (negative cost differences) compared to baseline expenditure budgets.

The right hand side of Figure 8 shows that households that rent tend to experience

weaker policy impacts (which are negative in 2020 and positive in 2050) than home owners if Switzerland switches to dynamic electricity tariffs. The latter seems relevant: The largest benefits of dynamic tariffs in the long run tend to accrue to home owners.

Figure A.1 in the Appendix reveals that, drilling down further into household composition, we find that among households with retirees as household heads, both single person households and couples are impacted by negative outcomes in 2020. Regional differences also exist, but more in terms how widely impacts are spread within regions than in terms of median impact magnitude (Figures A.3 and A.4 in the Appendix). Comparing BEV owners with households without BEVs (Figure A.5 in the Appendix) highlights the benefits of owning a BEV to take advantage of dynamic tariffs. This makes sense from the perspective of our analysis, since BEVs provide battery storage without additional investment costs.

3.3 Sensitivity of results to model assumptions

The numerical results in our ex-ante evaluation of transitioning from flat electricity tariffs to dynamic ones depends on a range of assumptions. One assumption is that of future electricity prices. The discussion in Figures 6–8 uses two different sets of assumptions about prices, and it becomes apparent that both magnitude and distribution of differences in direct cost depend crucially on this assumption. From the point of view of households, the uncertainty about future prices is difficult to handle. They will want to invest before prices materialize and may be inadequately equipped if they made the wrong investment decisions. Their goal has to be to invest so that they can live with the outcomes under both scenarios (with scenarios weighted according to their perceived probability). We observe that the investment decisions depend at least as much on average price levels as they do on tariff design (compare middle and bottom panels in Figures 2 and 3).

Another assumption we have to make is about the availability of BEVs for load flexibility. Are BEVs mostly standing at home and are they available to react flexibly to price differences? Or do people use them to drive to work and they are unavailable for most of the day? This question cannot be answered on a per-household level based on the model and data employed here. The results presented above are based on the assumption that BEVs are *not* available during working hours (*commute* variant of BEV availability). To illustrate the importance of the question, we rerun the model with BEVs being plugged in at home when they are not being driven (*wfh* variant of BEV availability). The results in Figure 9 reveal that the difference from Fig. 7 is not dramatic but visible, particularly for the high-price scenario. Our scenarios assume that all households either have their BEVs available during working hours or not. In reality, different households will experience different situations with regard to BEV availability during the day. In particular, more affluent households may own several BEVs, increasing their chances of at least one of them being available to provide load flexibility at home during the day.

A third dimension that may be hard to fathom is the actual rationality of households and investors in building infrastructure. The transaction cost and misalignment of incentives in interactions between tenants and landlords is an example where optimal investments may not be achieved. Our standard formulation of the model, in scenario *flexible+invest*, assumes that landlords have all the incentives in place to invest in cost-saving infrastructure so that tenants enjoy an optimized energy system in the building where they live. In reality, transparency about energy costs on the

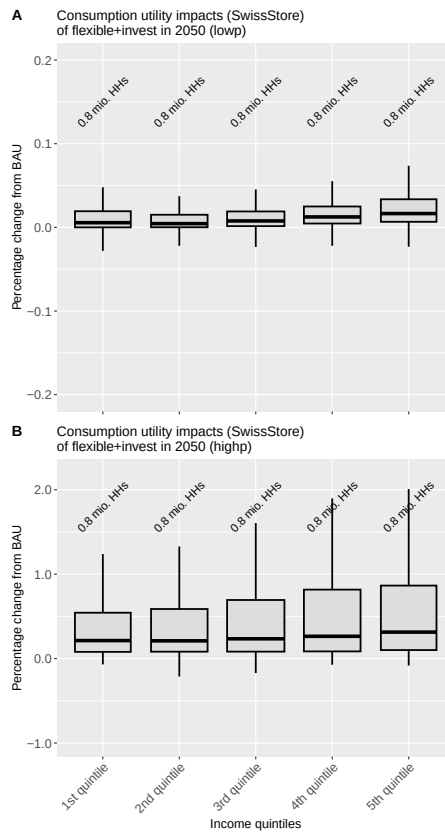


Figure 9: Left: Household specific outcomes for the *flexible+invest* case (*wfh* variant of BEV availability) in the *lowp* scenario for 2050 (panel A), and the *highp* scenario (panel B). Panels show relative cost reductions (negative cost differences) compared to baseline expenditure budgets.

real estate market and constructs like ZEVs (Zusammenschluss zum Eigenverbrauch) may work towards reaching this situation, but transaction costs may still keep households from reaching the cost minimal energy systems assumed in this model under the flexible+invest scenario. Even if landlords and tenants reach an agreement under a ZEV or something similar, it remains uncertain how such an agreement would distribute the surplus of optimally investing in efficiency and flexibility among participants. Our base assumption is that they would split the surplus evenly between tenants and landlords. But other deals may produce different allocations of surplus, and if all the surplus goes to tenants, landlords will not have any strong incentive to make optimal investments. These considerations motivate the sensitivity analysis shown in Figure 10. The first three columns of the figure assume optimal investments in solar PV and batteries by landlords and tenants but different sharing of the net surplus from investments in efficiency and flexibility. The first column assumes all net surplus to go to landlords (ts0), the second column assumes a 50–50 split (the base case assumption underlying all previously shown results; ts50), and the third column assumes 100 percent of net surplus to go tenants (ts100). If all net surpluses go to tenants, landlords do not have a strict incentive to invest, and therefore the fourth column shows results where property owners only invest in solar PV and batteries if they live in the dwelling themselves (model settings flexible+owners_invest; ts100). The comparison of results reveals that basic trends and patterns in the distribution of policy impacts will not change in a fundamental way under the different assumptions about interactions between landlords and tenants. The most notable change is that for the lowest income quintile group, impacts of introducing dynamic tariffs are the most positive if net surplus of energy infrastructure investments goes to landlords for the low price scenario for 2050, and least positive for the high price scenario for 2050.

In summary, we find our results to strongly depend on the price scenarios for 2050, but less so on the availability of EVs for electricity price arbitrage during the day and on specific details about the interactions between tenants and landlords.

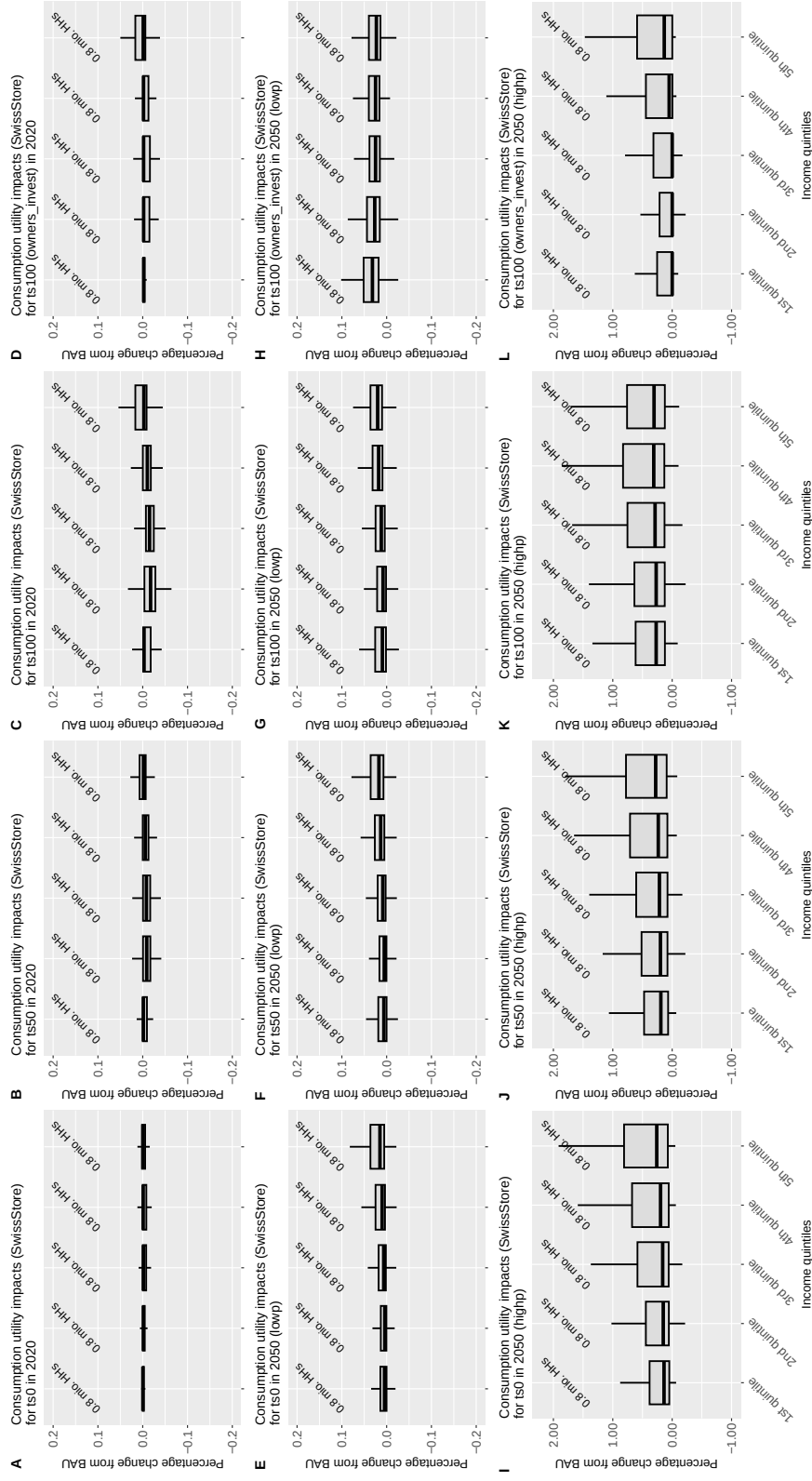


Figure 10: Distribution of changes in households' energy cost in 2020 (panels A–C) and 2050 in the lowp scenario (panels D–F) and in the highp scenario (panels G–I). The first three columns show results for the flexible+invest model settings where landlords invest optimally. The net surplus generated by these investments is shared between tenants and landlords in a 0–100 (panels A, E, and I), a 50–50 (panels B, F, and J), or a 100–0 (panels C, G, and K) split. The last column (panels D, H, and L) contains results for flexible+owners_invest model settings, where building owners invest in solar PV and batteries only if they live in their self-owned home and all surplus from investments in heat-pumps is going to tenants.

4 Conclusions

We use a model of rational agents' investments and use of flexibility in shifting electricity demand to analyze how a transition from flat electricity tariffs to dynamic tariffs would impact households in different parts of the Swiss population. We find that the distribution of impacts is mildly regressive in the short term and close to distributionally neutral in the long term if electricity prices stay low according to our low-price scenario for 2050. At the same time, the heterogeneity within household groups is considerable. We view BEV ownership as an exogenous decision, and we find that BEV owners profit more from dynamic tariffs than households without a BEV.

What do our results imply for policy making? We take the desirability of dynamic tariffs on system efficiency grounds as given. Our study then shows that, on the basis of equity, a straightforward implementation of dynamic tariffs in the short term may have slightly undesirable distribution of impacts across income levels. This result is gained through comparison of quite small differences that we can see in the graphic representation of our results, and one has to ask if these differences are relevant. We would argue that heterogeneity in impacts within groups should not be ignored, but that systematic differences between population groups other than income and home ownership are not of a magnitude that should be major drivers in our considerations about electricity tariff design and its complementary measures. It's the considerable heterogeneity of impacts within all population groups and the uncertain outcomes for single households that this implies can make the change in tariff design unpopular on its own. These drawbacks can be mitigated by accompanying measures that support negatively impacted households or by providing an option to opt out of the dynamic tariff. (This option is currently warranted by Swiss federal legislation.) Opting out would be attractive to all households that expect to experience increases in direct energy system costs in the short term. In the long-term, most customers should opt for dynamic tariffs according to our results.

But the best option may be to design and regulate for versions of dynamic tariffs that avoid systematic negative impacts of the new tariff design. Such designs could, e.g., leave customers electricity bill unchanged as long as household do not change their electricity consumption from what they use under a flat tariff, while giving full incentives for households to adjust their behavior according to spot market electricity prices. Winzer, Ramírez-Molina, Hirth, and Schlecht (2024) have suggested such a dynamic tariff model, in which utilities contract certain quantities and load profiles at a given price and only deviations from the profile are paid at the spot market electricity price. It remains to be explored how the design of the contracted profiles and their prices can be regulated so that the new contracts look equitable compared to existing flat tariffs.

Some caveats to our study design are in place. It should be noted that the disadvantage of renters and pensioners in our model is entirely due to the properties of the buildings and the income patterns of the households living therein. Our models do not consider the incomplete-information component of the principal-agent problem between house owners and their tenants or decreased motivation to invest by pensioners due to their approaching end of life. Another caveat of our study design is that our model does not take into account the effects that changing household behavior has on hourly spot market prices and thus on dynamic electricity tariffs themselves. Iterating between our model and FEM could yield additional insights, but it exceeds the scope of this study. But the strong influence of neighbouring countries' electricity price on

Swiss market prices for electricity makes us suspect that changes in household electricity consumption would have small effects only on the energy price component of residential electricity prices. Dynamic tariffs can also reduce the requirement for building out the electricity grid, since demand reductions at peak hours also reduce demand for grid capacity. This highlights that our study cannot capture some first order benefits of dynamic tariffs to the overall efficiency of the electricity system.

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CRedit author statement

Florian Landis: Conceptualization, Methodology, Software, Formal analysis, Resources, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization, Supervision, Project Administration, Funding acquisition.

Héctor Ramírez-Molina: Conceptualization, Methodology, Software, Formal analysis, Resources, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization.

Data statement

The models used herein run on different platforms and rely on data that are publicly available in some instances but confidential or governed by restricted access policies in others. For the above reasons, we feel it does not make sense to post model code, but we are willing to provide researchers that are interested in setting up (parts of) a similar model with the related code upon request.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Claude (AI) for programming assistance and DeepL for text styling. The authors reviewed and edited the output as needed and take full responsibility for the content of the published article.

References

Andersen, K. S., Wiese, C., Petrovic, S., & McKenna, R. (2020). Exploring the role of households' hurdle rates and demand elasticities in meeting danish energy-savings target. *Energy Policy*, 146, 111785. Retrieved from <https://www.sciencedirect.com/science/article/>

- pii/S030142152030505X doi: <https://doi.org/10.1016/j.enpol.2020.111785>
- Bollinger, B., Gillingham, K., & Kirkpatrick, A. J. (2025, January). *Valuing solar subsidies* (Working Paper No. 33368). National Bureau of Economic Research. Retrieved from <http://www.nber.org/papers/w33368> doi: 10.3386/w33368
- Boogen, N., & Winzer, C. (2024, November). *Households' Willingness to Curtail Electricity Usage During Winter Shortages - A Field Experiment* [SSRN Scholarly Paper]. Rochester, NY: Social Science Research Network. Retrieved 2026-01-29, from <https://papers.ssrn.com/abstract=5037079> doi: 10.2139/ssrn.5037079
- Borenstein, S. (2005, July). The Long-Run Efficiency of Real-Time Electricity Pricing. *The Energy Journal*, 26(3), 93–116. Retrieved 2026-01-29, from <https://doi.org/10.5547/ISSN0195-6574-EJ-Vol26-No3-5> doi: 10.5547/ISSN0195-6574-EJ-Vol26-No3-5
- Borenstein, S. (2013, March). Effective and Equitable Adoption of Opt-In Residential Dynamic Electricity Pricing. *Review of Industrial Organization*, 42(2), 127–160. Retrieved 2026-01-29, from <https://doi.org/10.1007/s11151-012-9367-3> doi: 10.1007/s11151-012-9367-3
- Bundesamt für Energie BFE. (2020, November). *ENERGIEPERSPEKTIVEN 2050+ Zusammenfassung der wichtigsten Ergebnisse* (Tech. Rep.). Bundesamt für Energie BFE. Retrieved 2026-01-10, from <https://www.bfe.admin.ch/bfe/de/home/politik/energieperspektiven-2050-plus.html>
- Bundesamt für Energie BFE. (2021). *Schweizerische Elektrizitäts-Statistik 2020* (Tech. Rep.). Bundesamt für Energie BFE. Retrieved 2026-03-09, from <https://www.bfe.admin.ch/bfe/en/home/supply/statistics-and-geodata/energy-statistics/electricity-statistics.html/>
- Bundesamt für Energie BFE. (2026). *Einspeisevergütung*. Retrieved 2026-05-27, from <https://www.bfe.admin.ch/bfe/de/home/foerderung/erneuerbare-energien/einspeiseverguetung.html>
- Bundesamt für Statistik. (2014). *Steckbrief Haushaltserhebung: Neues Gewich-tungsmodell, Resultate 2000-2003 und Studie zur Altersvorsorge*. Retrieved 2015-02-04, from http://www.bfs.admin.ch/bfs/portal/de/index/infothek/erhebungen_quellen/blank/blank/habe/01.html
- Darudi, A., Savelsberg, J., & Schlecht, I. (2024). *Thrive in sunshine, brace for thunder: Least-cost robust power system investments under political shocks* (Working Paper). Kiel, Hamburg: ZBW - Leibniz Information Centre for Economics. Retrieved 2026-01-10, from <https://www.econstor.eu/handle/10419/306555>
- De Groote, O., & Verboven, F. (2019, June). Subsidies and time discounting in new technology adoption: Evidence from solar photovoltaic systems. *American Economic Review*, 109(6), 2137–72. Retrieved from <https://www.aeaweb.org/articles?id=10.1257/aer.20161343> doi: 10.1257/aer.20161343
- Elcom. (2024). *Strompreise Schweiz*. Retrieved 2026-01-10, from <https://www.strompreis.elcom.admin.ch/>
- Farsi, M., & Weber, S. (2024). *Swiss Household Energy Demand Survey: Past ex-*

- periences and new perspectives* (Working Paper No. 24-06). IRENE Working Paper. Retrieved 2025-02-27, from <https://www.econstor.eu/handle/10419/306529>
- Faruqui, A., & Sergici, S. (2010, October). Household response to dynamic pricing of electricity: a survey of 15 experiments. *Journal of Regulatory Economics*, 38(2), 193–225. Retrieved 2026-01-29, from <https://doi.org/10.1007/s11149-010-9127-y> doi: 10.1007/s11149-010-9127-y
- García-Gusano, D., Espegren, K., Lind, A., & Kirkengen, M. (2016). The role of the discount rates in energy systems optimisation models. *Renewable and Sustainable Energy Reviews*, 59, 56–72. Retrieved from <https://www.sciencedirect.com/science/article/pii/S1364032116000253> doi: <https://doi.org/10.1016/j.rser.2015.12.359>
- Haq, G., & Weiss, M. (2018). Time preference and consumer discount rates - insights for accelerating the adoption of efficient energy and transport technologies. *Technological Forecasting and Social Change*, 137, 76–88. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0040162517311988> doi: <https://doi.org/10.1016/j.techfore.2018.06.045>
- Hausman, J. A. (1979). Individual discount rates and the purchase and utilization of energy-using durables. *The Bell Journal of Economics*, 10(1), 33–54. Retrieved 2026-08-20, from <http://www.jstor.org/stable/3003318>
- Heeter, J., Sekar, A., Fekete, E., Shah, M., & Cook, J. (2021). *Affordable and accessible solar for all: Barriers, solutions, and on-site adoption potential* [Other]. doi: 10.2172/1820098
- Krzywinski, M., & Altman, N. (2014, February). Visualizing samples with box plots. *Nature Methods*, 11(2), 119–120. Retrieved 2024-01-10, from <https://www.nature.com/articles/nmeth.2813> (Number: 2) doi: 10.1038/nmeth.2813
- Lawrance, E. C. (1991). Poverty and the rate of time preference: Evidence from panel data. *Journal of Political Economy*, 99(1), 54–77. Retrieved 2026-08-20, from <http://www.jstor.org/stable/2937712>
- Löffler, K. (2021, sep). Social discounting, social costs of carbon, and their use in energy system models. *Environmental Research Letters*, 16(10), 104005. Retrieved from <https://doi.org/10.1088/1748-9326/ac228a> doi: 10.1088/1748-9326/ac228a
- Marcucci, A. (2022). *Assumptions CROSS scenarios v2022-09*. Retrieved 2026-01-11, from <https://sweet-cross.ch/data/assumption-cross-scenarios/2022-09-30/>
- Marcucci, A., Dujardin, J., Heinisch, V., Panos, E., & Yilmaz, S. (2022). *CROSS Scenarios and Drivers Definition* (Tech. Rep. No. CROSS-v2022-09). SWEET-CROSS. Retrieved 2026-01-11, from https://sweet-cross.ch/wp-content/uploads/2023/07/CROSS_scenarios.V2022-09_2023.07.19.pdf
- Mellot, A., Moretti, C., Tröndle, T., & Patt, A. (2024, June). Mitigating future winter electricity deficits: A case study from Switzerland. *Energy Conversion and Management*, 309, 118426. Retrieved 2026-08-11, from <https://www.sciencedirect.com/science/article/pii/S0196890424003674> doi: 10.1016/j.enconman.2024.118426
- Newell, R. G., & Siikamäki, J. (2015, May). Individual time preferences and energy efficiency. *American Economic Review*, 105(5), 196–200. Retrieved

- from <https://www.aeaweb.org/articles?id=10.1257/aer.p20151010> doi: 10.1257/aer.p20151010
- Nägeli, C., Jakob, M., Catenazzi, G., & Ostermeyer, Y. (2020). Policies to decarbonize the swiss residential building stock: An agent-based building stock modeling assessment. *Energy Policy*, 146, 111814. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0301421520305322> doi: <https://doi.org/10.1016/j.enpol.2020.111814>
- Ovaere, M., & Vergouwen, M. (2026, June). *Capacity Constraints and Electrification: Evidence from Residential Electricity Demand and Pricing* (Working paper). Ghent University, Faculty of Economics and Business Administration. Retrieved 2026-08-11, from <https://EconPapers.repec.org/RePEc:rug:rugwps:25/1115>
- Parajeles Herrera, M., & Hug, G. (2025, June). Modeling Charging Demand and Quantifying Flexibility Bounds for Large-Scale BEV Fleets. In *2025 IEEE Kiel PowerTech* (pp. 1–6). Retrieved 2026-08-11, from <https://ieeexplore.ieee.org/document/11180551> doi: 10.1109/PowerTech59965.2025.11180551
- Rinaldi, A., Yilmaz, S., Patel, M. K., & Parra, D. (2022, October). What adds more flexibility? An energy system analysis of storage, demand-side response, heating electrification, and distribution reinforcement. *Renewable and Sustainable Energy Reviews*, 167, 112696. Retrieved 2026-01-10, from <https://www.sciencedirect.com/science/article/pii/S1364032122005858> doi: 10.1016/j.rser.2022.112696
- Schlecht, I., Darudi, A., Holmer, D., Willemsen, S., Ladermann, A., Derichs, J., ... Schneeberger, S. (2025). *Bedarf an Energiespeichern in der Schweiz* (Final project report). Bundesamt für Energie BFE. Retrieved 2026-08-11, from <https://www.bfe.admin.ch/bfe/de/home/versorgung/stromversorgung/stromversorgungssicherheit.exturl.html/aHR0cHM6Ly9wdWJkYi5iZmUuYWRTaW4uY2gvZGUvchVibGljYX/Rpb24vZG93bmXvYWQvMTIzOTI=.html>
- Schlecht, I., Ramírez-Molina, H., & Darudi, A. (2025). *AGGREGATE – The value of aggregators in a flexible and decentralized Swiss energy system* (Final project report). Retrieved 2026-01-09, from <https://www.aramis.admin.ch/Texte/?ProjectID=51523>
- Schleich, J., Gassmann, X., Faure, C., & Meissner, T. (2016). Making the implicit explicit: A look inside the implicit discount rate. *Energy Policy*, 97, 321–331. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0301421516304050> doi: <https://doi.org/10.1016/j.enpol.2016.07.044>
- Schleich, J., Gassmann, X., Meissner, T., & Faure, C. (2023). Making the factors underlying the implicit discount rate tangible. *Energy Policy*, 177, 113563. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0301421523001489> doi: <https://doi.org/10.1016/j.enpol.2023.113563>
- Schweingruber, A. C., Epple, R., Oetliker, U., & Rochat, S. (2007). *Die Einkommens- und Verbrauchserhebung mit neuer Methode* (Tech. Rep. No. 343561). Neuchâtel: Federal Statistical Office (BFS). Retrieved 2026-01-09, from <https://dam-api.bfs.admin.ch/hub/api/dam/assets/343561/master>

- Steinbach, J., & Zamasz, K. (2017, 01). Discount rates for the evaluation of energy projects – rules and problems. *Scientific Papers of Silesian University of Technology. Organization and Management Series*, 2017, 571-584. doi: 10.29119/1641-3466.2017.101.42
- Streicher, K. N., Padey, P., Parra, D., Bürer, M. C., Schneider, S., & Patel, M. K. (2019, February). Analysis of space heating demand in the Swiss residential building stock: Element-based bottom-up model of archetype buildings. *Energy and Buildings*, 184, 300–322. Retrieved 2026-01-10, from <https://www.sciencedirect.com/science/article/pii/S0378778818325660> doi: 10.1016/j.enbuild.2018.12.011
- Torné, A., & Trutnevyte, E. (2024, February). Banning fossil fuel cars and boilers in Switzerland: Mitigation potential, justice, and the social structure of the vulnerable. *Energy Research & Social Science*, 108, 103377. Retrieved 2025-04-04, from <https://www.sciencedirect.com/science/article/pii/S2214629623004371> doi: 10.1016/j.erss.2023.103377
- Train, K. (1985). Discount rates in consumers' energy-related decisions: A review of the literature. *Energy*, 10(12), 1243-1253. Retrieved from <https://www.sciencedirect.com/science/article/pii/0360544285901355> doi: [https://doi.org/10.1016/0360-5442\(85\)90135-5](https://doi.org/10.1016/0360-5442(85)90135-5)
- Winzer, C., Ludwig, P., Auer, S., & Hlawatsch, A. (2023, January). *NETFLEX – Effiziente Netzentgelte für flexible Verbraucher* (Tech. Rep.). Zürich: Bundesamt für Energie BFE. Retrieved 2026-01-10, from <https://www.aramis.admin.ch/Texte/?ProjectID=44252>
- Winzer, C., Ramírez-Molina, H., Hirth, L., & Schlecht, I. (2024, December). Profile contracts for electricity retail customers. *Energy Policy*, 195, 114358. Retrieved 2026-01-09, from <https://www.sciencedirect.com/science/article/pii/S0301421524003781> doi: 10.1016/j.enpol.2024.114358

A Additional results

In Figures A.1–A.4 additional results show how impacts of introducing dynamic targets compare for additional household groupings besides the grouping according to income, home ownership, and age. And what is the contribution of owning a BEV to these outcomes? Figure A.5 shows that BEV owners profit more (lose less) from the switch from flat to dynamic tariffs in 2020 and, to a lesser extent, in 2050.

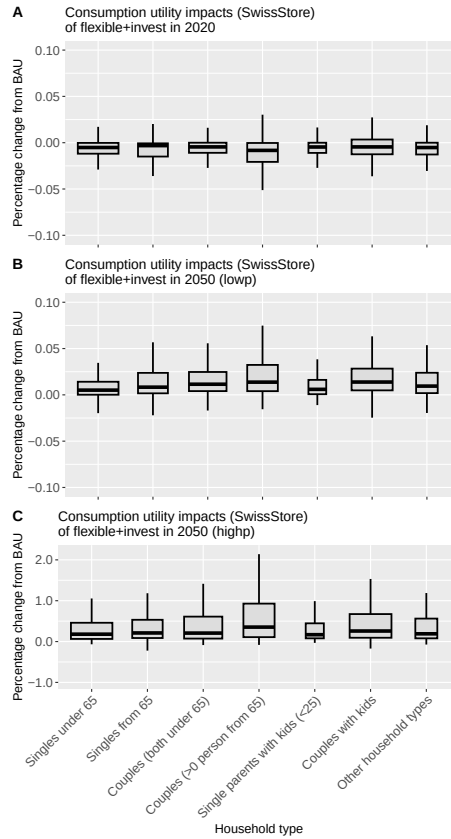


Figure A.1: Household specific outcomes differentiated with respect to household type in 2020 (panel A) and 2050 in a low price scenario (panel B) and in a high price scenario (panel C). Panels show relative cost reductions (negative cost differences) compared to baseline expenditure budgets.

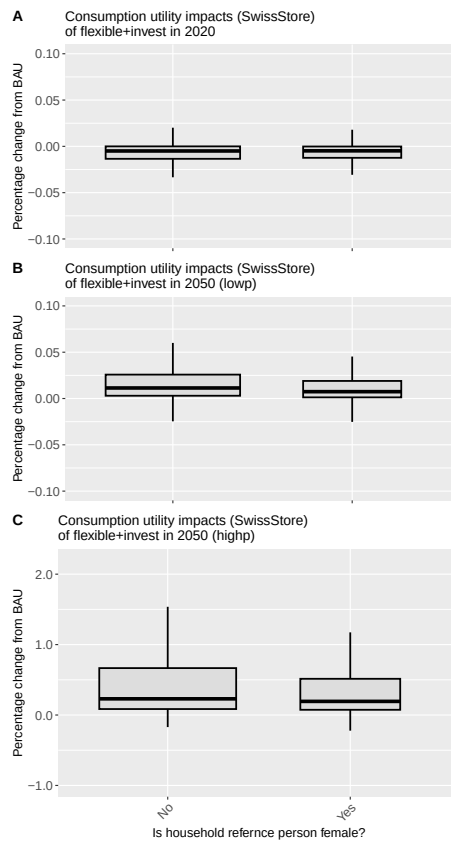


Figure A.2: Household specific outcomes differentiated with respect to gender in 2020 (panel A) and 2050 in a low price scenario (panel B) and in a high price scenario (panel C). Panels show relative cost reductions (negative cost differences) compared to baseline expenditure budgets.

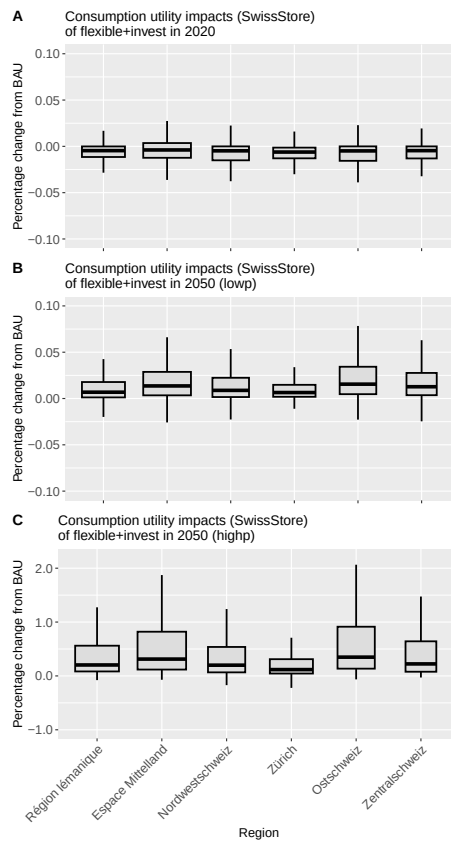


Figure A.3: Household specific outcomes differentiated with respect to region in 2020 (panel A) and 2050 in a low price scenario (panel B) and in a high price scenario (panel C). Panels show relative cost reductions (negative cost differences) compared to baseline expenditure budgets.

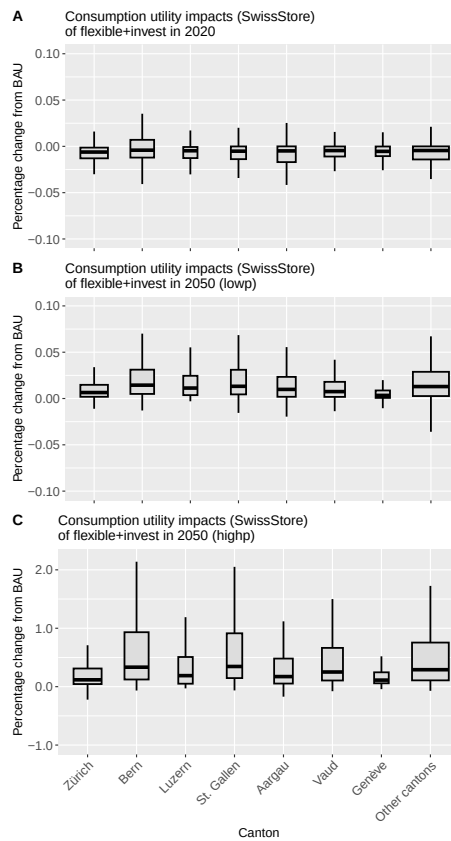


Figure A.4: Household specific outcomes differentiated with respect to canton in 2020 (panel A) and 2050 in a low price scenario (panel B) and in a high price scenario (panel C). Panels show relative cost reductions (negative cost differences) compared to baseline expenditure budgets.

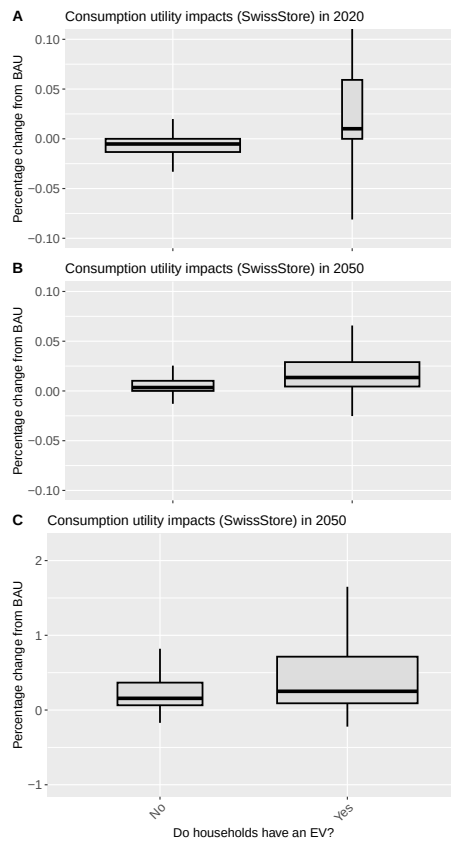


Figure A.5: Household specific outcomes differentiated with respect to BEV ownership in 2020 (panel A) and 2050 in a low price scenario (panel B) and in a high price scenario (panel C). Panels show relative cost reductions (negative cost differences) compared to baseline expenditure budgets.